Allison Casey

Del Mar, CA, 92014| allisonscasey@gmail.com |858-353-0656

EDUCATION

Texas Christian University Fort Worth, TX
B.S. Mechanical Engineering | Minor: Innovation and Entrepreneurship May 2028
John V. Roach Honors College | Cumulative GPA: 3.87
Relevant Coursework: Solid Mechanics I-II, Statics , Dynamics & Vibrations, Control Systems, Thermal Sciences, Circuit Analysis & Design I-II, Electronics, Applied Programming (MATLAB), Computer Science (Honors)
Affiliations: ASME, Theta Tau Professional Engineering Fraternity, Society of Women Engineers (SWE), Chi Omega

EXPERIENCE

R&D Engineering Intern - Spur Team May 2026-August 2026
Reflow Medical San Clemente, CA
- Performed device builds, dimensional inspections, bench testing, and simulated-use testing on retrievable peripheral and coronary stent systems, generating quantitative data analysis to support design verification
- Designed and built a portable artery model in SolidWorks for physician demonstrations and trade shows, integrating custom tubing, 3D-printed components, and a motorized water-circulation system
- Authored and revised engineering documentation, including test protocols, manufacturing procedures, work orders, and lab notebook studies in support of product development
- Collaborated cross-functionally with R&D, manufacturing, quality, marketing, and physicians, presenting findings through Excel data analysis and technical PowerPoints

Engineering Teaching Assistant and Engineering Clinic Staff August 2025-present
Texas Christian University, Department of Engineering Fort Worth, TX
- Selected as a sophomore to support 100+ students weekly through lab instruction, grading, and office hours; teach MATLAB and Autodesk Inventor and guide hands-on design projects

Private STEM Tutor January 2025-present
Math Around the Corner Fort Worth, TX
- Teach middle and high school students one-one in Algebra, Calculus, Physics, and Chemistry

PROJECTS

Illuminart: Sensory Lightbox
Rady Children's Hospital Helen Bernardy Center
- Designed and built an adaptive, interactive, light box to promote cognitive function for medically fragile children
- Integrated an easy-touch switch, brightness control, and ten light modes. Ensured devices met hospital spec requirements including portability, battery-power, and ease to sanitize.

Calla: Wearable Tech Bracelet
Texas Christian University Independent Research Project (Incoming, Fall 2026)
- Designing a health wearable bracelet to blend in as standard jewelry while tracking vital metrics for health, tailored specifically for women
- Research focuses on how physiological variances in female bodies influence sensor performance and data interpretation

Aircraft Structural Patch Optimization & Strain Analysis
Texas Christian University Engineering Design Project
- Designed a riveted aluminum repair patch for a damaged aircraft skin panel, reducing stress at the crack tip by 93% and cutting panel deflection by 67%
- Simulated the repair in Nastran FEA and verified the design on a physical test bench using strain gauges, matching computer predictions to physical test results within 7.7%

SKILLS

CAD & Simulation: SolidWorks, Autodesk Inventor, Autodesk Nastran (FEA), LTspice
Instrumentation & Testing: Strain Gauge, Oscilloscope, Digital Multimeter, Function Generator, DAQ System
Programming & Data Analysis: MATLAB, Java, Javascript, Arduino (C++), Python, MS Excel
Fabrication & Rapid Prototyping: 3D Printing (FDM/SLA), Laser Cutting, Precision Soldering, Metrology Inspection

AWARDS & PUBLICATIONS

- Rostam Reifschneider Endowed Award for Leadership, Innovation, and Entrepreneurship
- Published in Ellen Lupton's "Health Design Thinking: Volume 2"
- Top 3 National Finalist, Cooper Hewitt National High School Design Competition